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RESEARCH-ARTICLE

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Generative AI for Emotional and Cultural Engagement in Interactive Games

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Abstract

This study applies generative artificial intelligence (AI) to cultural interactive design through Chuling, an AI-driven music game based on Norman's emotional design model. By integrating multimodal interaction, the system investigates how generative AI enhances user emotion and cultural identity. Tests with 50 participants show that AI significantly improves engagement and cultural resonance. The findings highlight generative AI's role as a co-agent for emotional interaction and cultural narration, offering pathways for its application in public cultural spaces and digital media art.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms.

Keywords

Game Design, Generative AI, Emotional Design, Cultural Interactive Game

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1 INTRODUCTION

With digital transformation, public cultural spaces are shifting from static displays to immersive, personalized experiences [1]. Current interactive systems, however, rely on preset feedback, limiting real-time emotion and cultural resonance. Generative AI offers a solution by enabling adaptive, emotion-linked feedback beyond traditional procedural content generation [2]. While prior studies explore AI in visual generation, semantic interaction, and storytelling

[3], systematic integration into emotional design with empirical validation remains scarce.

This study applies Norman's three-level emotional design model [4] in developing Chuling, an AI-driven cultural music game combining color recognition, AI visual generation, motion capture, radar sensing, and LLM-based feedback. A user study with 50 participants demonstrates that Generative AI significantly enhances behavioral and reflective experiences compared to non-AI interaction. The contributions are threefold: (1) proposing an AI-emotional design integration pathway, (2) implementing a multimodal cultural game prototype, and (3) empirically validating AI's role in advancing cultural interaction.

2 THEORETICAL FOUNDATIONS AND RELATED WORK

Generative AI has evolved from procedural content generation to a central role in interactive gameplay [5], with applications in character modeling, narrative, and player feedback. Frameworks such as SCENECRAFT integrate LLMs with player data for dynamic storytelling [6], while Cox and Ooi emphasize context-adaptive NPC dialogues [7]. Studies at Aalborg University show that LLM-based NPCs enhance presence and emotional engagement [8]. These findings inspire the integration of generative AI into multimodal cultural game interactions.

In cultural contexts, AI-driven chatbots improve visitor engagement and understanding through personalized guidance [9]. Pont highlights AI curation's role in reflection and ethics [10], while Spennemann notes its potential to foster cultural identification in heritage exhibitions [11]. However, current applications remain limited to content delivery and guided tours, lacking emotional resonance.

In summary, existing research shows that while AI enhances interaction, narrative, and cultural engagement, it is seldom integrated systematically into emotional design. Prior work emphasizes information provision over emotional feedback, and calls for prioritizing long-term cultural values beyond technical novelty. Building on these insights, this study adopts Norman's three-level emotional design model to couple generative AI with multimodal interaction, advancing a pathway of sensory appeal-emotional feedback-cultural identity.

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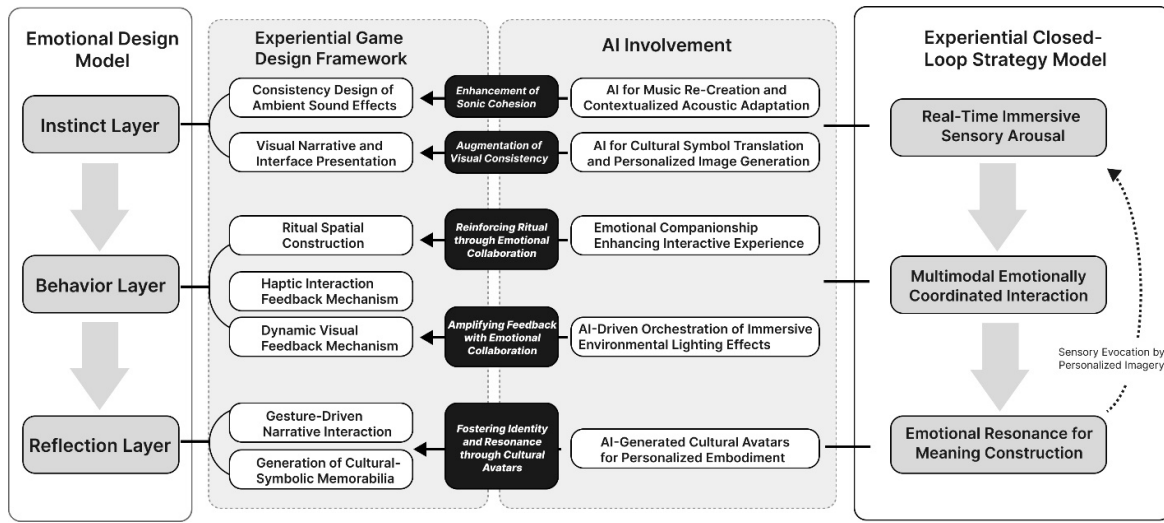


Figure 1: Research on Multidimensional Fusion Pathways for Generative AI in Empowering Emotional Game Experiences.

3 INTERACTION DESIGN

The Emotional Design framework models user emotions in games through Norman’s three layers—instinctive, behavioral, and reflective—emphasizing positive affect via sensory appeal, operational feedback, and meaning-making^[4]. Generative AI introduces new opportunities for affective design.

3.1 Game Device Design

3.1.1 Visceral Level — Immediate Sensory Appeal. The instinct layer shapes players’ initial impressions through visual, auditory, and aesthetic stimuli^[12]. During this process, AI co-agent generate cultural scenarios and personalized deity avatars based on Chu culture elements and player attire. They also create cohesive environmental sound effects and performance sound effects synchronized with main music segments to establish immersive cultural ambiance. By dynamically aligning sensory outputs with individual player contexts, the AI actively participates in crafting immersive experiences that resonate culturally^[13].

3.1.2 Behavioral Level — AI-Supported Multimodal Emotional Feedback. The behavioral layer facilitates real-time interaction and emotional feedback^[14]. Players use gesture-based controls to enter the Performance Zone and strike rhythmic targets with specialized mallets. Missed notes trigger vibrotactile feedback. Unlike template-based systems, the AI co-agent analyzes real-time performance data—such as strike accuracy and error rates—to deliver personalized verbal and textual feedback at set thresholds (30%, 60%, 90%). It identifies structural error patterns (e.g., persistent rhythm deviations) and provides proactive guidance. Through contextual adaptation, the system varies feedback, avatars, and narratives across sessions, framing the AI as a creative collaborator rather than a parametric tool, thereby promoting co-creation and engagement.

3.1.3 Reflective Level — AI-Generated Cultural Avatars and Emotional Extension. The reflection layer supports meaning-making and

sustained emotional engagement^{[4][15]}. At the narrative climax, the AI generates deity avatars based on players’ clothing colors, linking gameplay actions to cultural metaphors through avatar mapping. Personalized avatars and collectible cards featuring verses from Chu Ci extend engagement beyond real-time interaction. Each player’s unique signature cards foster distinctiveness and encourage cultural reflection. Within this framework, the AI strengthens cultural identity, transforming transient gameplay into enduring cultural resonance and emotional connections.

Through this co-agent framework, generative AI operates across sensory, behavioral, and reflective layers, facilitating immersive cultural visuals, emotion-aware interactions, and reflective identity formation. The resulting sensory-exchange-reflection loop demonstrates AI’s capacity to co-create cultural meaning with players, providing a transferable design paradigm for interactive cultural heritage games (Figure 1).

3.2 System Implementation and Technical Workflow

Chuling integrates radar sensing, rhythm interaction, and AI-driven personalization within a unified architecture (Figure 2). The system consists of three modules: (1) camera-based color recognition with ControlNet-guided Stable Diffusion for deity avatar generation, (2) gesture and rhythm detection for synchronized performance feedback, and (3) millimeter-wave radar enabling touch-free multiplayer interaction to enhance immersion and cultural resonance.

3.2.1 Implementation. The prototype was implemented in Unity 2023.7, integrating PyTorch and Transformers for model inference, Stable Diffusion v2.1 for image generation, and GPT-4o for natural language interaction. It ran on a workstation with an NVIDIA RTX 4090 GPU (24 GB VRAM), a HIKVISION 4K camera, a Pavo LiDAR, and a 4K projector for immersive rendering. While optimized for high-performance hardware, the framework remains scalable through lightweight or cloud-based deployment on VR/AR and mobile platforms.

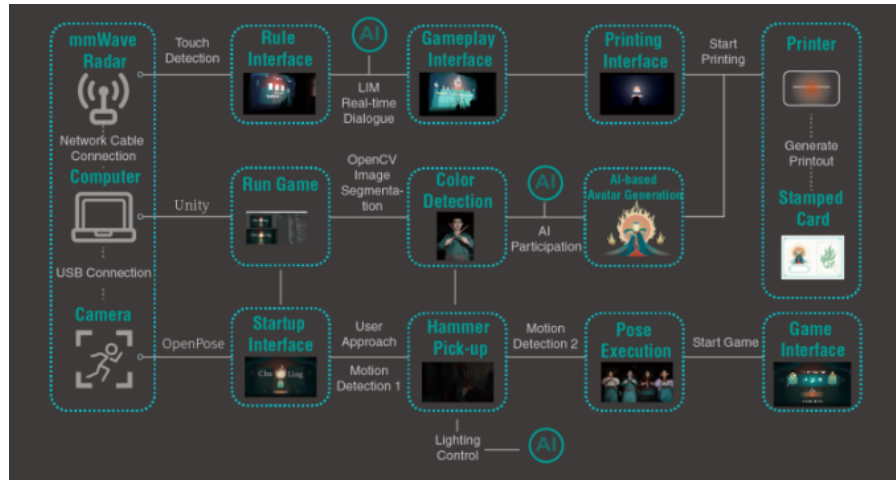


Figure 2: System Architecture of Chuling.

The prototype, developed in Unity, uses C# scripts to synchronize rhythm mechanics and player input. AI modules were integrated for personalization: real-time avatar generation via ControlNet-guided Stable Diffusion; milestone-based LLM feedback for voice and NPC dialogues; and multimodal interaction via OpenPose and millimeter-wave radar. The system ran on a workstation with an NVIDIA RTX 4090 GPU, a HIKVISION 4K camera, a Pavo LiDAR, and a 4K projector, remaining scalable for cloud or lightweight deployment.

3.2.2 Potential Application Scenarios. Chuling is designed for public cultural spaces like museums, heritage sites, and digital art exhibitions. In a typical museum setting, it allows visitors to interact with traditional culture digitally. The AI-generated avatars and personalized cultural cards serve as both entertainment and educational tools, improving cultural understanding and retention. With radar-based multi-user support, it suits group activities in cultural festivals or educational programs. The flexible architecture also enables adaptation to VR/AR environments for remote immersive experiences.

4 SYSTEM EVALUATION

This study examines the effects of AI co-agent in cultural rhythm games on sensory appeal, immersion, and cultural resonance, including variations across players' cultural backgrounds and prior familiarity, and its impact on recognition, memorability, and emotional engagement with cultural elements.

4.1 Sampling and participants

Fifty participants (aged 18–45) were recruited via stratified random sampling by professional (professionals vs. non-professionals) and cultural background (Chinese vs. foreign). Each group included 10 foreign- and 15 Chinese-background participants. Participants originated from China (n = 30), Pakistan (n = 6), Nigeria (n = 4), the US (n = 3), Japan (n = 2), South Africa (n = 2), Brazil (n = 2), and Germany (n = 1). Session order (AI vs. non-AI) was counterbalanced.

The predominance of Chinese participants limits cross-cultural generalizability, with non-China participants included for exploratory assessment of AI effects on reflective-level cultural engagement.

4.2 Operationalization

Participants experienced two conditions: the AI-driven version of ChuLing(Condition A), featuring personalized dialogues, adaptive card generation, and avatars, and the non-AI version of ChuLing(Condition B), which retained core rhythm gameplay but excluded these AI features. Experiments took place in a simulated museum with a large projection screen, high-precision cameras, and radar sensors for motion tracking; lighting and sound were calibrated for immersion, with assistants ensuring system stability. Engagement was measured via a Likert-scale questionnaire e.g., O'Brien & Toms, and cultural resonance was assessed through semi-structured interviews coded for recognition, emotional connection, and narrative appreciation.

4.3 Data Collection and Analysis

1) Quantitative Measures:

User engagement was evaluated using a three-layer scale adapted from O'Brien & Toms (2010). The structure included: (1) Instinct Layer assessed visual and auditory attractiveness and initial engagement. (2) Behavior Layer - Interaction Feedback (4 items) evaluated controllability, clarity and consistency of system responses, emotional encouragement, and immersion. (3) Reflection Layer - Cultural Identification (4 items) measured cultural understanding, emotional connection, intention for repeated engagement, and recommendation.

All items used a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Dimension scores were averaged, and internal consistency was assessed with Cronbach's α . The results of AI VS Non-AI are summarized in Table 2 to Table 3, differences in cultural background are summarized in Table 4, presenting the means and standard deviations for each dimension.

Table 1: Composition of participants.

Sample grouping	Cultural Background	number of people	Total
Specialized groups	Chinese background	15	25
	Foreign background	10	
Non-specialist group	Chinese background	15	25
	Foreign background	10	

Table 2: Professional Group and Non-Professional Group – User Evaluation (Mean ± SD).

Group	Level	AI(Mean ± SD)	Non-AI(Mean ± SD)
Professional Group	Instinct Layer: Sensory Appeal	4.12 (0.68)	3.95 (0.70)
	Behavior Layer: Interaction	4.05 (0.71)	3.56 (0.75)
	Feedback		
	Reflection Layer: Cultural Identification	4.28 (0.65)	3.76 (0.67)
Non-Professional Group	Instinct Layer: Sensory Appeal	4.08 (0.66)	3.92 (0.72)
	Behavior Layer: Interaction	3.89 (0.82)	3.49 (0.71)
	Feedback		
	Reflection Layer: Cultural Identification	4.22 (0.70)	3.69 (0.63)

Table 3: AI vs Non-AI – Overall Comparison (Mean ± SD, Mean Difference, Cohen’s d, p-value).

LEVEL	Condition A (Mean ± SD)	Condition B (Mean ± SD)	Mean Difference (A–B)	Estimated Effect Size (Cohen’s d)	Difference Description
Instinct Layer: Sensory Appeal	4.12 (0.68)	3.95 (0.70)	0.17	0.24	Small difference, not significant
Behavior Layer: Interaction	4.05 (0.72)	3.56 (0.75)	0.49	0.66	Condition A significantly higher than Condition B (p < 0.01)
Feedback					
Reflection Layer: Cultural Identification	4.28 (0.69)	3.76 (0.67)	0.52	0.77	Condition A significantly higher than Condition B (p < 0.01)

Table 4: Cross-cultural comparison of Chinese and non-China participants (AI condition).

LEVEL	Chinese (n = 30)M±SD	Foreign (n = 20)M±SD	t	p	Cohen’s d
Instinct Layer: Sensory Appeal	4.03 ± 0.10	3.91 ± 0.03	9.93	<0 .001	1.38
Behavior Layer: Interaction	3.81 ± 0.23	3.54 ± 0.02	10.24	<0 .001	1.3
Feedback					
Reflection Layer: Cultural Identification	4.05 ± 0.26	3.73 ± 0.02	10.98	<0 .001	1.38

Data analysis was conducted using mean, standard deviation (SD), significance (p-value), and effect size (Cohen’s d). Mean scores were interpreted as 1.0–2.4 = dissatisfied, 2.5–3.4 = neutral, 3.5–4.4 = satisfied, and 4.5–5.0 = very satisfied. SD indicated consistency of responses: ≤0.5 = highly consistent, 0.6–1.0 = relatively consistent, >1.0 = highly variable. Significance was set at p < 0.05 for meaningful differences between AI and non-AI conditions. Cohen’s d

was used to assess effect magnitude: 0.2 = small, 0.5 = medium, and ≥0.8 = large.

2) Qualitative Measures:

Interview Supplement: Semi-structured interviews with a subset of participants further explored experiences across the three

layers. Thematic analysis identified sub-themes such as visual/auditory appeal, interaction fluency and immersion, and cultural resonance and dissemination intent, providing qualitative support to the scale findings.

Instinct Layer – Sensory Appeal: Most respondents ($n=38$) found visuals and lighting immersive, and two-thirds ($n=34$) noted motivating rhythmic audio. Sensory appeal was observed across groups, with higher emphasis among Chinese participants. **Behavior Layer – Interaction Feedback:** Some users ($n=4$) reported occasional unresponsiveness, but over half ($n=43$) experienced strong immersion during smooth interactions. Chinese participants reported slightly higher flow levels. **Reflection Layer – Cultural Identification:** Cultural resonance was stronger among Chinese participants, while non-Chinese respondents showed greater curiosity in cultural elements. Overall, 70% expressed sharing intent, with Chinese users emphasizing cultural transmission and non-Chinese users focusing on learning.

3) Synthesis of Results

Measurement Findings: As shown in Table 3, Condition A scored higher than Condition B across all layers. The instinct layer difference was non significant ($M = 0.17$, $d = 0.24$), while AI significantly improved behavioral ($M = 0.49$, $d = 0.66$, $p < 0.01$) and reflective outcomes ($M = 0.52$, $d = 0.77$, $p < 0.01$). Independent samples t-tests further revealed that Chinese participants scored higher than Non-China participants on sensory appeal, interaction feedback, and cultural identity ($ps < 0.001$, Cohen's $d > 1.30$).

Interview Findings: Participants noted that visuals, lighting, and drum beats enhanced instinctive engagement. Usability issues occasionally reduced fluency, but real-time feedback fostered immersion. Chinese participants emphasized cultural resonance and personal memories, while non-China participants highlighted curiosity and memorability. Both groups recognized cultural value and expressed willingness to share the experience.

4.4 Discussion

In summary, AI co-creation significantly enhances player experience in cultural rhythm games compared to the non-AI version. While Chinese-background participants scored higher overall, AI-assisted scores exceeded non-AI scores for both groups, indicating improved interactive feedback and cultural identity. Participants noted occasional system recognition issues, and sensory immersion gains were less visually apparent. Qualitative findings showed that visuals, lighting, and rhythmic audio promoted instinctive engagement, while real-time guidance and adaptive dialogue deepened immersion. The generative nature of the AI ensured that each interaction was unique, even for repeat players. This unique cultural embodiment makes every experimenter think it's interesting. Chinese participants emphasized cultural resonance, and non-China participants exhibited curiosity and memorability. Both groups acknowledged cultural value and willingness to share experiences, demonstrating that AI-driven avatars and dynamic content support cross-cultural understanding and transmission.

Overall, this experiment has the following limitations: 1. Although the samples have non-China cultural backgrounds, the overall sample size is small, and cross-cultural adaptability is still in the exploratory stage. Future studies should expand the sample

size for further exploration; 2. Although quantitative and qualitative experimental research were conducted, the study primarily relied on subjective evaluation indicators from experimenters. Future research should incorporate multimodal affect recognition and physiological measurement methods to validate emotional responses; 3. There is a lack of measurement studies before and after user usage; 4. The research mainly focuses on self-comparison between AI and non-AI versions of ChuLing, lacking comparative studies with other generative AI cultural systems, which limits its relative effectiveness evaluation and universality verification.

5 CONCLUSION AND FUTURE WORK

This study developed an emotional cultural rhythm game using generative AI and validated it on the Chuling platform. The AI co-agent improved behavioral experience and reflective engagement via real-time interaction feedback and cultural recognition, though gains in initial sensory appeal were limited. The study integrates Norman's affective design framework with generative AI, forming a pathway across sensory appeal, adaptive interaction, and cultural meaning construction. Practically, the Chuling platform is scalable for large exhibitions, VR/AR, and virtual heritage applications. Future work will expand sample diversity, incorporate physiological and multimodal measures, explore ethical considerations, and compare with other generative AI cultural systems to assess generalizability. Investigations into long-term learning, memory retention, and cultural co-creation, along with context-aware and culturally adaptive prompts, may further enhance engagement, cross-cultural adaptability, and the educational value of digital cultural experiences.

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